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Short Communication

Revisiting the blind mind: Still no evidence for sensory visual imagery in individuals with aphantasia

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ABSTRACT

The inability to visualise was given the name aphantasia in 2015 by Zeman and colleagues. In 2018 we published research showing that fifteen individuals who self-identified as having aphantasia also demonstrated a lack of sensory visual imagery when undergoing the binocular rivalry imagery paradigm, suggesting more than just a metacognitive difference. Here we update these findings with over fifty participants with aphantasia and show that there is evidence for a lack of sensory imagery in aphantasia. How the binocular rivalry paradigm scores relate to the vividness of visual imagery questionnaire (VVIQ) and how aphantasia can be confirmed is discussed.

In 2015 Zeman and colleagues published a paper about a group of individuals who reported the inability to visualise, in other words, they had a 'blind mind's eye' (Zeman et al., 2015), much like individuals described in Galton's seminal work (Galton, 1880, 1883). Zeman et al. (2015) gave this experience a name: congenital aphantasia. Since then much research and interest has burst forth about aphantasia, both in the scientific arena, and from the general population (Bainbridge et al., 2021; Clemens, 2018; Dawes et al., 2020; de Vito and Bartolomeo, 2016; Grinnell, 2016; Jacobs et al., 2018; Keogh et al., 2021; Milton et al., 2021; Ross, 2016; Wicken et al., 2021; Zeman et al., 2016).

Much of the research on aphantasia has centered around investigating whether having aphantasia impacts other cognitive abilities. Thus far research has demonstrated that individuals with aphantasia can perform lab-based psychophysical and intelligence-type clinical visual working memory tasks similarly and sometimes better than individuals with imagery, potentially through using non-visual imagery strategies to compensate (Keogh et al., 2021; Milton et al., 2021; Pounder et al., 2022). There is evidence that the ability to create attention templates may be affected in individuals with aphantasia (Keogh and Pearson, 2021; Monzel et al., 2021), however, see also (Cabbai et al., 2023). The most convincing and potentially important cognitive function that may be impacted by having aphantasia is autobiographical memory. Research has demonstrated that individuals with aphantasia report reduced memories of the past (Dawes et al., 2020; Watkins, 2018; Zeman et al., 2020), as well as evidence of less vivid episodic memories of the past when assessed using autobiographical interviews (Dawes et al., 2022; Milton et al., 2021).

The field of aphantasia is still in its infancy and the veracity of self-reported aphantasia is still questioned by some due to the personal nature of visual imagery and heavy reliance on introspection and subjective reports (de Vito and Bartolomeo, 2016), although it is worth pointing out that this argument can apply to any internal experience that relies on self-report. With the rapid increase of online studies and a renewed interest in aphantasia, an important question keeps surfacing, how can we best identify or confirm aphantasia?

In 2018 we published a study using the binocular rivalry imagery technique (Pearson, 2014; Pearson et al., 2008) showing that self-identified individuals with aphantasia (N = 15) also lacked evidence of visual imagery when measured using this paradigm (Keogh and Pearson, 2018), supporting their introspective reports. However, this was a relatively small sample, and we could not report on the likelihood that self-identified individuals with aphantasia would also show an absence of sensory imagery using this technique.

Here we extend our previous findings by presenting the data we have collected at our laboratory over the years since this paper was published, again using the binocular rivalry paradigm, in a larger group of self-identified individuals with aphantasia (N = 55), including the 15 participants from the 2018 study, plus participants who have participated in other lab-based published and unpublished studies (Kay et al., 2022; Keogh and Pearson, 2021; Keogh et al., 2021; Wicken et al., 2021). The binocular rivalry imagery task was the same as that reported in our original 2018 paper (see Fig. 1). After undergoing an eye-dominance

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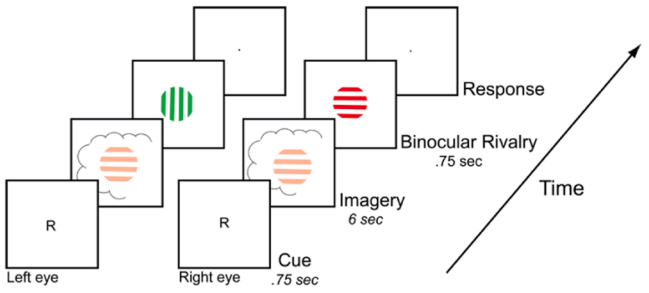


Fig. 1. *Binocular rivalry imagery paradigm timeline.* Binocular rivalry was achieved using red-green anaglyph glasses. For ease of understanding the timeline shows what each eye is presented through the filters (e.g. left eye = green filter and right eye = red-filter) however these images were physically shown in the same location on the computer screen. During the imagery task participants were cued to imagine either a red horizontal or green vertical Gabor patch (letter R and G, respectively) on a black background for six seconds. Following this they were presented with a binocular rivalry display comprised of a red-horizontal and green-vertical Gabor patch and participants were asked to indicate which image was dominant (green = 1, mixed = 2 and red = 3).

calibration task participants completed the binocular rivalry imagery paradigm. They were instructed to imagine either a red-horizontal or green-vertical grating (letter cues R and G respectively) for six seconds. After this, they were presented with a brief (750 ms) binocular rivalry display and they were asked to indicate which image was dominant. The percentage of trials in which the dominance of binocular rivalry was the same as the cue prime (i.e. imagine red and see red) was measured (known as percent primed). As with previous work, we interpret higher priming scores to indicate stronger sensory imagery. This is because a physically presented weak image also primes or facilitates subsequent binocular rivalry (Brascamp et al., 2007; Pearson et al., 2008), and imagery shares a number of other priming characteristics with perception (such as orientation (Keogh and Pearson, 2017; Pearson et al., 2008) and location specificity (Bergmann et al., 2016)), suggesting that visual imagery has sensory characteristics.

In line with our previous findings, the priming scores (our measure of imagery strength) in those with aphantasia (N = 55, see teal violin plot in Fig. 2A, mean = 52.49, SD = 8.51) were significantly lower than the general populations (N = 209, see pink violin plot, mean = 58.83, SD = 11.65), Mann-Whitney t-test used due to violation of normality and differences in sample sizes (W = 3640.5, $p < .001$). We also ran a Bayesian Mann-Whitney analysis and this provided strong evidence that the aphantasia group had lower priming scores than the control group (W = 3640.5, $BF_{10} = 14.26$). Next, we assessed whether individuals with aphantasia, on average, showed any evidence of above-chance (50%) priming. A one-sample t-test (Wilcoxon signed-rank test used for violation of normality) showed that, as a group, individuals with aphantasia did not have priming that was significantly different from chance (V = 837, $p = .284$). This was further supported by a Bayesian analysis (Wilcoxon signed-rank analysis for violation of normality) which provided evidence in favour of the null regarding whether individual's with aphantasia binocular rivalry priming scores were greater than chance ($BF_{01} = 21.46$). Finally, we used the new mean priming score we obtained from the 55 individuals with aphantaisa and compared this to the bootstrap analysis run in the 2018 paper (see Fig. 2B, white bars). This bootstrap analysis involved randomly sampling 15 participants from the general population and averaging their priming scores 1000 times. The number of times the mean scores of these random samples were the same or lower than the aphantasia mean priming score was then assessed (see the black dashed line and white bars in Fig. 2B). We found that out of 1000 iterations, the new mean priming score (52.49%) was equal to or less than 4 of the 1000 iterations, or $p = .004$. As this previous bootstrap analysis used simulated samples of 15 to match our sample of those with aphantasia at the time, we wanted to update this to match the new sample size of 55 (see grey bars in Fig. 2B). When this was done 0 out of 1000 iterations of a 55-person sample had a mean equal to or lower than the aphantasia sample (the lowest bootstrapped mean in this case was 55%). Taken together, these data replicate and extend our 2018 paper, supporting a lack of sensory imagery in self-identified aphantasia.

When our previous paper was published there was no consistent process used to assess aphantasia, other than self-identification so we did not exclude any participants and thus we did not set any cut-off criteria for our participants (this is also the case in Fig. 2A). With the rise of interest in aphantasia the question of how best to confirm aphantasia has been raised frequently. To date the vividness of visual imagery questionnaire (VVIQ (Marks, 1973) has been most frequently used to 'confirm aphantasia' with two main approaches used in the

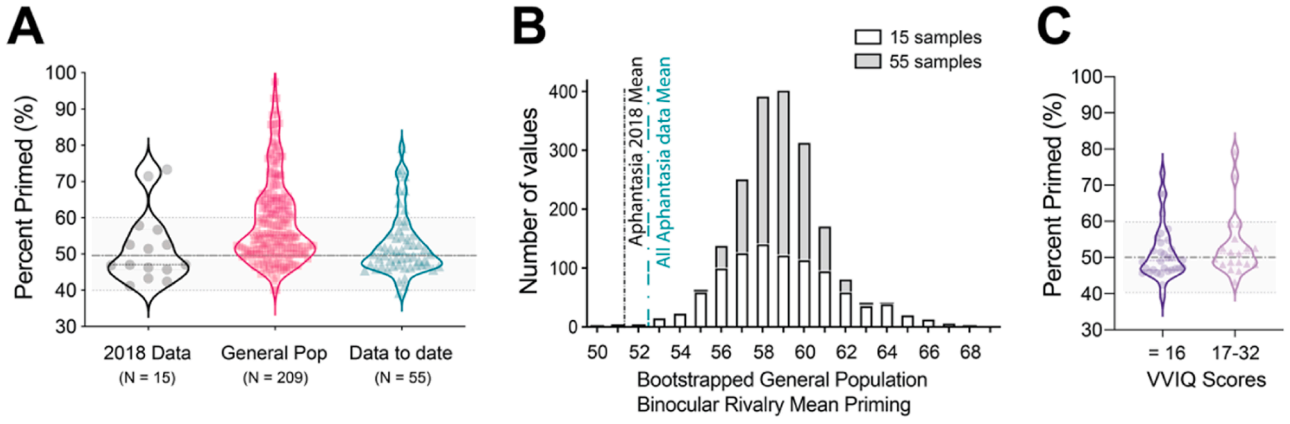


Fig. 2. *Binocular rivalry priming data.* ***A.*** Data shows binocular rivalry priming scores from the 2018 paper for aphantasia (left grey-scale violin plot) and general population data (middle pink violin plot). The rightmost teal violin plot shows data from all individuals with aphantasia tested on the binocular rivalry paradigm to date (including those from the 2018 paper). **B.** Frequency histograms (stacked) for the bootstrapping analysis' from the general population data (Bins = 1) – white bars represent the same bootstrapping as the 2018 paper for the 15 sample and the grey bars represent the new bootstrap analysis drawing 55 participants to match the new aphantasia group sample size. For the analysis 15 or 55 subjects priming scores were randomly chosen, averaged, then returned to the main pool of general population subjects. This was done 1000 times each for the 2 bootstrap analysis'. Data shows the distribution of the mean of N = 15 for 1000 iterations (white bars) and mean of N = 55 for 1000 iterations (grey bars). The aphantasia groups mean from the 2018 paper is shown on the far left (grey dotted line), while the new data is represented with a teal dotted line. **C.** Violin plots show binocular rivalry data scores for individuals with aphantasia with scores below 32 on the VVIQ (N = 51) split into two groups: VVIQ = 16 (N = 32, left dark purple violin plot) and VVIQ = 17–32 (N = 19, right light purple violin plot). Each individual data point in the violin plots represents an individual participant, and lines on violin plots represent medians and quartiles.

papers that investigate aphantasia. One approach is to include anyone who scores a 32 or lower on the VVIQ, which coincides with the highest score given by self-identified individuals with aphantasia in the original Zeman paper (Zeman et al., 2015), equivalent to the imagery being either vague and dim (rating of 2) or no image at all (rating of 1) on the VVIQ. The second way researchers have labelled participants is by separating them into those who have 'aphantasia' (those who score 16 on VVIQ) and those with 'weak or limited visual imagery' (self--identified individuals with aphantasia who score between 17–32 on the VVIQ) (Zeman et al., 2020). For this reason, we decided to use these two forms of classification to assess whether these different classifications would be reflected in differences in binocular rivalry priming.

Using the first criteria (scores less than 32 on the VVIQ), four participants in thissample were removed due to VVIQ scores greater than 32 (33, 34, 37 and 39) and of these 4 participants 2 had priming scores above 60% (mean = 57.08, SD = 13.09). The new group of individuals with aphantasia comprised of 51 participants with VVIQ scores of 32 or below and had a mean binocular rivalry priming score of 52.07% (SD = 8.09), which equates to a mean that was greater than or equal to 3 out of the 1000 iterations for the first bootstrap analysis or a $p = .003$ (see Fig. 2B white bars).

Using the second criteria that has been previously used to separate individuals with aphantasia, we split the data into two groups 'self-identified individuals with aphantasia with a VVIQ score of 16′ and 'self-identified individuals with aphantasia with VVIQ scores of 17–32′ (see Fig. 2C dark and light purple plots respectively). The individuals with aphantasia who scored 16 on the VVIQ (left dark purple violin plot in Fig. 2C, N = 32, Mean = 51.32, SD = 7.57) did have a mean priming score that was numerically lower than the individuals who scored 17–32 on the VVIQ (right light purple violin plot in Fig. 2C, N = 19, Mean = 53.33, SD = 8.97) however, this difference was not significant (Mann-Whitney independent t-test W = 247.5, $p = .14$). This lack of an effect was also partially supported by a Bayesian analysis with anecdotal evidence for a null effect $BF_{01} = 1.65$. Finally, priming in both groups was not significantly different to chance (16: V = 241, $p = .44$; 17–32: V = 121.00, $p = .15$) and a Bayesian analysis provided moderate evidence for the null effect for both groups (16: $BF_{01} = 10.93$; 17–32: $BF_{01} = 4.68$).

Finally, due to an increase in online testing, many researchers have asked the question, what proportion of people who report having no visual imagery would likewise also show a lack of sensory visual imagery? While all participants here attended the laboratory in person, they all contacted us via email or signing up to our online database and thus we think their results are likely representative of the participants who participate in online studies. For the participants who scored 32 or lower on the VVIQ, 12% of those (6/51) had a priming score greater than or equal to 60% (which is 10% above chance and also roughly the average priming score found in many binocular rivalry studies measuring visual imagery). We also analysed the proportion of individuals who scored above 60% priming by separating them into those with a VVIQ score of 16 and those with VVIQ scores of 17–32. For those with a score of 16 on the VVIQ, 4 out of 32 (12.5%) had scores above 60% (priming scores: 62.2%,67.65%, 67.5%,73.33% - mean: 67.67%), while for participants with a VVIQ score of 17–32, 2 out of 19 (10.5%) had scores above 60% (priming scores: 72.42%,79.21% - mean: 75.82%). To summarise, the proportion of people who report no imagery (VVIQ<33 +self-identification) and likewise show priming < 60% equates to about 88%, while for VVIQ16 +self-identification equates to around 87% and VVIQ17–32 +self-identification equates to about 89%. However, it should be noted these two subgroups have smaller sample sizes and hence should be revisited in time with larger datasets.

In sum, we extended our previous findings (Keogh and Pearson, 2018) to a larger group of individuals with aphantasia. Specifically, as a whole, this group of individuals did not show evidence of sensory visual imagery when measured using the binocular rivalry paradigm. One interesting finding however is that approximately 12% of self-identified individuals with aphantasia had scores above 60% on the binocular rivalry priming paradigm, so it may be possible that these individuals have some form of 'unconscious visual imagery' that leads to priming or perhaps the differences reside in the metacognitive process of self-evaluation of imagery vividness. It could be interesting for future research to investigate whether these individuals who report having aphantasia, but prime binocular rivalry above 60%, likewise might also show evidence of visual imagery using other objective measures (e.g. pupillometry, skin conductance or even decoding in neuroimaging).

These data also suggest that dividing individuals with aphantasia into 2 sub-groups may not provide much further explanatory power when conducting analyses. This may be because individuals who have aphantasia but provide some 2 ratings on the VVIQ may be interpreting 'dim and vague' to mean something completely different to someone with visual imagery. For example, some of the participants who attend our laboratory and provide some 'dim and vague' ratings on the VVIQ report that while they don't see an image they may get a 'feeling' of the spatial layout of a scene and thus a rating of 1 does not quite capture their experience and that is why they provide a 2. This lines up with other work that finds that spatial imagery is not reduced in aphantasia (Bainbridge et al., 2021; Dawes et al., 2020). Future research should aim to develop qualitative questionnaires that better reflect the phenomenological experience of individuals with aphantasia in addition to more behavioural and objective indexes of aphantasia.

CRediT authorship contribution statement

Keogh Rebecca: Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Pearson Joel:** Funding acquisition, Resources, Writing – review & editing, Conceptualization, Methodology, Supervision.

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References

Bainbridge, W.A., Pounder, Z., Eardley, A.F., Baker, C.I., 2021. Quantifying aphantasia through drawing: Those without visual imagery show deficits in object but not spatial memory. Cortex 135, 159–172. https://doi.org/10.1016/j.cortex.2020.11.014.

Bergmann, J., Genc, E., Kohler, A., Singer, W., Pearson, J., 2016. Smaller primary visual cortex is associated with stronger, but less precise mental imagery. Cereb. Cortex 26 (9), 3838–3850. https://doi.org/10.1093/cercor/bhv186.

Brascamp, J.W., Knapen, T.H.J., Kanai, R., van Ee, R., van den Berg, A.V., 2007. Flash suppression and flash facilitation in binocular rivalry. J. Vis. 7 (12) doi:Artn 12 10.1167/7.12.12.

Cabbai, G., Brown, C.R.H., Dance, C., Simner, J., Forster, S., 2023. Mental imagery and visual attentional templates: A dissociation. Cortex 169, 259–278. https://doi.org/10.1016/j.cortex.2023.09.014.

Clemens, A. (2018). When the Mind's Eye Is Blind. *Scientific American*. Retrieved from ⟨https://www.scientificamerican.com/article/when-the-minds-eye-is-blind1/⟩.

Dawes, A.J., Keogh, R., Andrillon, T., Pearson, J., 2020. A cognitive profile of multi-sensory imagery, memory and dreaming in aphantasia. Sci. Rep. 10 (1), 10022 https://doi.org/10.1038/s41598-020-65705-7.

Dawes, A.J., Keogh, R., Robuck, S., Pearson, J., 2022. Memories with a blind mind: Remembering the past and imagining the future with aphantasia. Cognition 227, 105192. https://doi.org/10.1016/j.cognition.2022.105192.

Galton, F., 1880. Statistics of Mental Imagery. Mind 5, 301–318.

Galton, F. (1883). Inquiries into human faculty and its development. *Macmillan*.

Grinnell, D. (2016). My mind's eye is blind – so what's going on in my brain? *New Scientist*. Retrieved from ⟨https://www.newscientist.com/article/2083706-my-minds-eye-is-blind-so-whats-going-on-in-my-brain/⟩.

Jacobs, C., Schwarzkopf, D.S., Silvanto, J., 2018. Visual working memory performance in aphantasia. Cortex 105, 61–73. https://doi.org/10.1016/j.cortex.2017.10.014.

Kay, L., Keogh, R., Andrillon, T., Pearson, J., 2022. The pupillary light response as a physiological index of aphantasia, sensory and phenomenological imagery strength. Elife 11. https://doi.org/10.7554/eLife.72484.

Keogh, R., Pearson, J., 2017. The perceptual and phenomenal capacity of mental imagery. Cognition 162, 124–132. https://doi.org/10.1016/j.cognition.2017.02.004.

Keogh, R., Pearson, J., 2018. The blind mind: No sensory visual imagery in aphantasia. Cortex 105, 53–60. https://doi.org/10.1016/j.cortex.2017.10.012.

Keogh, R., Pearson, J., 2021. Attention driven phantom vision: measuring the sensory strength of attentional templates and their relation to visual mental imagery and aphantasia. Philos. Trans. R. Soc. Lond. B Biol. Sci. 376 (1817), 20190688. https://doi.org/10.1098/rstb.2019.0688.

Keogh, R., Wicken, M., Pearson, J., 2021. Visual working memory in aphantasia: Retained accuracy and capacity with a different strategy. Cortex 143, 237–253. https://doi.org/10.1016/j.cortex.2021.07.012.

Marks, D.F. (1973). Visual imagery differences in the recall of pictures. *Br J Psychol*, *64* (1), 17–24. Retrieved from ⟨http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?cmd=Retrieve&db=PubMed&dopt=Citation&list_uids=4742442⟩.

Milton, F., Fulford, J., Dance, C., Gaddum, J., Heuerman-Williamson, B., Jones, K., Zeman, A., 2021. Behavioral and neural signatures of visual imagery vividness extremes: aphantasia versus hyperphantasia. Cereb. Cortex Commun. 2 (2), tgab035 https://doi.org/10.1093/texcom/tgab035.

Monzel, M., Keidel, K., Reuter, M., 2021. Imagine, and you will find - Lack of attentional guidance through visual imagery in aphantasics. Atten. Percept. Psychophys. 83 (6), 2486–2497. https://doi.org/10.3758/s13414-021-02307-z.

Pearson, J., 2014. New Directions in Mental-Imagery Research: The Binocular-Rivalry Technique and Decoding fMRI Patterns. Curr. Dir. Psychol. Sci. 23 (3), 178–183. https://doi.org/10.1177/0963721414532287.

Pearson, J., Clifford, C.W., Tong, F., 2008. The functional impact of mental imagery on conscious perception. Curr. Biol. 18 (13), 982–986. https://doi.org/10.1016/j.cub.2008.05.048.

Pounder, Z., Jacob, J., Evans, S., Loveday, C., Eardley, A.F., Silvanto, J., 2022. Only minimal differences between individuals with congenital aphantasia and those with typical imagery on neuropsychological tasks that involve imagery. Cortex 148, 180–192. https://doi.org/10.1016/j.cortex.2021.12.010.

Ross, B. (2016). Aphantasia: How It Feels To Be Blind in your Mind. Retrieved from ⟨https://www.facebook.com/notes/blake-ross/aphantasia-how-it-feels-to-be-blind-in-your-mind/10156834777480504/⟩.

de Vito, S., Bartolomeo, P., 2016. Refusing to imagine? On the possibility of psychogenic aphantasia. A commentary on Zeman et al. (2015). Cortex 74, 334–335. https://doi.org/10.1016/j.cortex.2015.06.013.

Watkins, N.W., 2018. A)phantasia and severely deficient autobiographical memory: Scientific and personal perspectives. Cortex 105, 41–52. https://doi.org/10.1016/j.cortex.2017.10.010.

Wicken, M., Keogh, R., Pearson, J., 2021. The critical role of mental imagery in human emotion: insights from fear-based imagery and aphantasia. Proc. Biol. Sci. 288 (1946), 20210267. https://doi.org/10.1098/rspb.2021.0267.

Zeman, A., Dewar, M., Della Sala, S., 2015. Lives without imagery - Congenital aphantasia. Cortex 73, 378–380. https://doi.org/10.1016/j.cortex.2015.05.019.

Zeman, A., Dewar, M., Della Sala, S., 2016. Reflections on aphantasia. Cortex 74, 336–337. https://doi.org/10.1016/j.cortex.2015.08.015.

Zeman, A., Milton, F., Della Sala, S., Dewar, M., Frayling, T., Gaddum, J., Winlove, C., 2020. Phantasia – the psychological significance of lifelong visual imagery vividness extremes. Cortex.